

Wasserstein-2 Error Analysis of Riemannian Flow Matching:

A Rigorous Analytical Framework via Jacobi Field Theory

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Abstract

This project develops a rigorous analytical framework bounding the end-to-end Wasserstein-2 (W_2) estimation error of a flow matching model operating on a Riemannian manifold $(\mathcal{M}, \mathbf{g})$ of constant sectional curvature K . We prove five interconnected results. First, we derive the explicit ODE governing optimal-transport flow matching on $(\mathcal{M}, \mathbf{g})$ via the Brenier–McCann theorem and show that the true singularity $t \rightarrow 1^-$ is the spatial Lipschitz blow-up $\|\nabla_x u_t\| \lesssim C_\kappa/(1-t)$, not the speed along trajectories. Second, we adapt the Alekseev–Gröbner variation-of-constants formula to the Riemannian setting, expressing the trajectory deviation as $d_g(\tilde{X}_T, X_T) \leq \int_0^T \|D\Phi_{T,s}\|_{\text{vel}} |\delta_s| ds$, and clarify that curvature enters only through the flow Jacobian, not through the linearized ODE directly. Third, we deploy Jacobi field theory with initial conditions $J(0) = 0$, $D_t J(0) = \eta$ to derive the normalized velocity-perturbation amplification factor $A_\kappa(T) = \sinh(\kappa T)/(\kappa T)$, proving that the negative curvature ($K < 0$, hyperbolic geometry) exponentially amplifies truncation error as $A_\kappa(T) \sim e^{\kappa T}/(2\kappa T)$, while the positive curvature ($K > 0$) suppresses it. Fourth, we combine these results into a four-term master inequality

$$W_2(\hat{\mu}, \mu_1) \leq C_3 \tau + C_0 \sqrt{d} N^{-1/d} + A_\kappa(1-\tau) \left[C_1 \sqrt{\frac{d}{N}} \log \frac{1}{\tau} + \frac{C_2}{p} h^p (\tau^{-p} - 1) \right],$$

bounding the total error W_2 as a joint function of intrinsic dimension d , sample size N , curvature bound κ , solver step size h , and early-stopping time τ . Fifth, we analytically optimize over τ : under the condition $W_2(\mu_0, \mu_1) > C_1 \sqrt{d/N} + C_2 h^p$, an interior optimum τ^* exists in $(0, 1)$ and satisfies the exact balance condition $C_3 = A'_\kappa(1-\tau^*)F(\tau^*) + A_\kappa(1-\tau^*)[C_1 \sqrt{d/N}/\tau^* + C_2 h^p (\tau^*)^{-p-1}]$, with closed-form approximations in the field-error and numerical error that dominated regimes.

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1 Curvature-Dependent Differential Equations

1.1 Setting and Notation

Let $(\mathcal{M}, \mathbf{g})$ be a complete connected Riemannian manifold of dimension d with constant sectional curvature $K \in \mathbb{R}$. Write $\exp_x : T_x \mathcal{M} \rightarrow \mathcal{M}$ for the Riemannian exponential map and $\log_x : \mathcal{M} \rightarrow T_x \mathcal{M}$ for its local inverse. Let $\mathcal{P}_2(\mathcal{M})$ be the space of Borel probability measures on $\mathcal{M}$ with finite second moment, metrized by

$$W_2(\mu, \nu)^2 = \inf_{\gamma \in \Gamma(\mu, \nu)} \int_{\mathcal{M} \times \mathcal{M}} d(x, y)^2 d\gamma(x, y), \quad (1)$$

where $\Gamma(\mu, \nu)$ is the set of couplings and $d(\cdot, \cdot)$ the geodesic distance.

Key notation. We use $\tau \in (0, 1)$ for the *early stopping parameter* and set $T = 1 - \tau$ for the *effective integration horizon*. Integration runs over $[0, T] = [0, 1 - \tau]$; the interval $[T, 1] = [1 - \tau, 1]$ is skipped, contributing the stopping residual E_{stop} . This convention is used consistently throughout.

1.2 Derivation of the Flow-Matching ODE on $(\mathcal{M}, \mathbf{g})$

Let $\mu_0, \mu_1 \in \mathcal{P}_2(\mathcal{M})$ with μ_0 be absolutely continuous with respect to Riemannian volume measure. By Brenier–McCann’s theorem on manifolds, the unique optimal transport map $T : \mathcal{M} \rightarrow \mathcal{M}$ from μ_0 to μ_1 has the form $T(x) = \exp_x(-\nabla\varphi(x))$ for a c -convex potential φ . Define the *displacement interpolant*

$$T_t(x) = \exp_x(t \cdot \log_x(T(x))), \quad \mu_t = (T_t)_\# \mu_0. \quad (2)$$

Derivation of the ODE. Fix x_0 with $x_1 = T(x_0)$. The curve $\gamma(t) = T_t(x_0)$ is the unique constant-speed geodesic from x_0 to x_1 , parametrized as

$$\dot{\gamma}(t) = \frac{d}{dt} \exp_{x_0}(t \cdot v_0) = d \exp_{x_0}(tv_0) \cdot v_0 =: v_t, \quad (3)$$

where $v_0 = \log_{x_0}(x_1) \in T_{x_0} \mathcal{M}$ and $d \exp$ is the differential of the exponential map. Since γ is a geodesic, $D_t \dot{\gamma} = 0$, giving the *geodesic ODE*

$$D_t v_t = 0, \quad v_t = \dot{\gamma}(t). \quad (4)$$

The velocity field $u_t : \mathcal{M} \rightarrow T\mathcal{M}$ is defined by $u_t(\gamma(t)) = \dot{\gamma}(t)$; integrating over μ_0 and differentiating the push-forward relation yields the continuity equation

$$\partial_t \mu_t + \operatorname{div}(\mu_t u_t) = 0. \quad (5)$$

Solving (4) along $\gamma(t) = T_t(x_0)$: since γ is a constant-speed geodesic of speed $s_0 = |v_0| = d(x_0, x_1)$, the unit tangent vector $\hat{e}(t) = \dot{\gamma}(t)/s_0$ is parallel transported along γ (i.e. $D_t \hat{e} = 0$, a consequence of (4)). For any $t \in [0, 1)$, the sub-arc $\gamma|_{[t, 1]}$ is itself a geodesic of length $(1 - t)s_0$ from $\gamma(t)$ to x_1 , with unit tangent $\hat{e}(t)$ at its base. Therefore,

$$\log_{\gamma(t)}(x_1) = (1 - t) s_0 \cdot \hat{e}(t) = (1 - t) d(x_0, x_1) \cdot \hat{e}(t), \quad (6)$$

while $\dot{\gamma}(t) = s_0 \cdot \hat{e}(t) = d(x_0, x_1) \cdot \hat{e}(t)$. Dividing (6) by $(1 - t)$ and comparing:

$$\frac{1}{1 - t} \log_{\gamma(t)}(x_1) = d(x_0, x_1) \cdot \hat{e}(t) = \dot{\gamma}(t).$$

This gives the **explicit conditional velocity field**

$$u_t(x \mid x_0, x_1) = \frac{1}{1 - t} \log_{T_t(x_0)}(x_1). \quad (7)$$

1.3 Asymptotic Blow-up as $t \rightarrow 1^-$

Proposition 1.1 (Velocity Blow-up: conditional boundedness vs marginal singularity). *Under (7):*

(i) **L^2 boundedness along the trajectories.** *The conditional speed $|u_t(x \mid x_0, x_1)| = d(x_0, x_1)$ is t -independent. Hence $\|u_t\|_{L^2(\mu_t)} = W_2(\mu_0, \mu_1) < \infty$.*

(ii) **Spatial Lipschitz / gradient blow-up.** *The spatial Jacobian of u_t satisfies*

$$\|\nabla_x u_t(x \mid x_1)\|_{\text{op}} \lesssim \frac{C_\kappa}{1 - t} \quad \text{as } t \rightarrow 1^-, \quad (8)$$

where C_κ depends on the curvature κ and the cut-locus geometry. This is the true source of singularity: the velocity field becomes increasingly stiff in space as $t \rightarrow 1^-$.

Proof. Part (i). By (7): $u_t(x \mid x_1) = \frac{1}{1 - t} \log_x(x_1)$. Using $|\log_{T_t(x_0)}(x_1)|_{\mathbf{g}} = (1 - t)d(x_0, x_1)$ gives $|u_t| = d(x_0, x_1)$; integrating over the optimal coupling yields $\|u_t\|_{L^2} = W_2(\mu_0, \mu_1)$.

Part (ii). Differentiating $u_t(x \mid x_1) = \frac{1}{1 - t} \log_x(x_1)$ with respect to x in normal coordinates centered at x_1 :

$$\nabla_x u_t(x \mid x_1) = \frac{1}{1 - t} \nabla_x \log_x(x_1).$$

By the Gauss lemma and curvature comparison, $\|\nabla_x \log_x(x_1)\|_{\text{op}} \leq C_\kappa$ uniformly away from the cut locus. Hence $\|\nabla_x u_t\| \lesssim C_\kappa/(1 - t)$, which diverges as $t \rightarrow 1^-$. This Lipschitz blow-up is what makes the ODE stiff: local truncation errors of a p -th order solver scale as $h^{p+1} \|D_t^{p+1} u_t\| \lesssim h^{p+1} (1 - t)^{-(p+1)}$ (by composing $p + 1$ spatial derivatives each contributing a factor $(1 - t)^{-1}$). $\square$

Remark 1.2. Note the distinction: the *speed* $|u_t| = d(x_0, x_1)$ along the optimal trajectory is bounded, while the *spatial gradient* $\|\nabla_x u_t\| \sim (1-t)^{-1}$ diverges. Early stopping at $\tau > 0$ limits the integration to $t \in [0, 1-\tau]$, where $\|\nabla_x u_t\| \leq C_\kappa/\tau < \infty$.

1.4 Curvature-Modified ODE via Jacobi Fields

On $(\mathcal{M}, \mathbf{g})$ with constant sectional curvature K , the deviation $J(t)$ of a one-parameter geodesic family satisfies the *Jacobi equation*

$$D_t^2 J + K \cdot J = 0. \quad (9)$$

For initial data $J(0) = J_0$, $D_t J(0) = J'_0$, explicit solutions are:

$$K = -\kappa^2 : \quad J(t) = J_0 \cosh(\kappa t) + \frac{J'_0}{\kappa} \sinh(\kappa t), \quad (10)$$

$$K = 0 : \quad J(t) = J_0 + J'_0 t, \quad (11)$$

$$K = +\kappa^2 : \quad J(t) = J_0 \cos(\kappa t) + \frac{J'_0}{\kappa} \sin(\kappa t). \quad (12)$$

These solutions underpin the amplification factor $A(\kappa, T)$ in Section 4, where $T = 1 - \tau$.

2 Riemannian Alekseev–Gröbner Formula

2.1 Classical Alekseev–Gröbner Formula

Given $\dot{x} = f(x, t)$ and $\dot{y} = f(y, t) + \delta(y, t)$ in $\mathbb{R}^n$, with flows Φ^f, Φ^g :

$$\Phi_t^g(y_0) - \Phi_t^f(x_0) = \int_0^t D_x \Phi_{t,s}^f(\Phi_s^g(y_0)) \cdot \delta(\Phi_s^g(y_0), s) ds. \quad (13)$$

2.2 Lipschitz Bound on the Riemannian Exponential Map

Lemma 2.1 (Exponential-map Lipschitz bound). *Let $(\mathcal{M}, \mathbf{g})$ have sectional curvature $|K| \leq \kappa^2$ and injectivity radius $\iota > 0$. For $v, w \in T_x \mathcal{M}$ with $|v|, |w| \leq r < \min(\iota, \pi/(2\kappa))$,*

$$d(\exp_x v, \exp_x w) \leq L(\kappa, r) |v - w|, \quad L(\kappa, r) = \frac{\sinh(\kappa r)}{\kappa r} \geq 1. \quad (14)$$

Proof. Let $\sigma(s) = \exp_x((1-s)v + sw)$, $s \in [0, 1]$, a curve on $\mathcal{M}$. Its tangent satisfies $|\dot{\sigma}(s)| \leq \|d\exp_x((1-s)v + sw)\|_{\text{op}} |w - v|$. By the Rauch comparison theorem, $\|d\exp_x(u)\|_{\text{op}} \leq \sinh(\kappa |u|)/(\kappa |u|)$ for $|u| \leq r$. Integrating over $s \in [0, 1]$ and bounding $|(1-s)v + sw| \leq r$:

$$d(\exp_x v, \exp_x w) \leq \int_0^1 |\dot{\sigma}| ds \leq \frac{\sinh(\kappa r)}{\kappa r} |v - w|. \quad \square$$

2.3 Riemannian Alekseev–Gröbner: Direct Integral Form

Theorem 2.2 (Riemannian Alekseev–Gröbner). *Let $\tilde{X}_t$ (exact trajectory of u_t) and X_t (approximate trajectory of $\tilde{u}_t = u_t + \delta_t$) start from the same point. Then for $T = 1 - \tau$:*

$$d_g(\tilde{X}_T, X_T) \leq \int_0^T \|D\Phi_{T,s}\|_{\text{vel}} \cdot \left| \delta_s(\tilde{X}_s) \right|_{\mathbf{g}} ds, \quad (15)$$

where $D\Phi_{T,s}$ is the differential of the exact flow map $\Phi_{T,s} : \mathcal{M} \rightarrow \mathcal{M}$, and $\|D\Phi_{T,s}\|_{\text{vel}}$ denotes the operator norm restricted to velocity perturbation channels (i.e. perturbations of the form $J(0) = 0$, $D_t J(0) = \eta$). By Proposition 3.1:

$$\|D\Phi_{T,s}\|_{\text{vel}} \leq A_\kappa(T - s). \quad (16)$$

Note: for initial-position perturbations ($J(0) \neq 0$), the operator norm is bounded by $\cosh(\kappa(T - s)) \geq A_\kappa(T - s)$; the bound (16) applies because the field errors δ_s act as velocity perturbations at each time s .

Proof. Flow differential approach (normal coordinates). Let $\Phi_{t,s} : \mathcal{M} \rightarrow \mathcal{M}$ be the exact flow map of u_t (i.e. $\Phi_{t,s}(x)$ is the position at time t of the trajectory starting from x at time s). Working in normal coordinates centered at $\tilde{X}_s$, the variation-of-constants formula for the perturbed ODE gives

$$\tilde{X}_T - X_T = \int_0^T D\Phi_{T,s}(\tilde{X}_s) \delta_s(\tilde{X}_s) ds + O(\|\delta\|^2), \quad (17)$$

where $D\Phi_{T,s}(x) : T_x \mathcal{M} \rightarrow T_{\Phi_{T,s}(x)} \mathcal{M}$ is the differential of the flow map (the *Riemannian flow Jacobian*).

Remark on the linearization. For a general first-order flow $\dot{X} = u_t(X)$, the perturbation equation for $\xi_t = \tilde{X}_t - X_t$ (in local coordinates) is $\dot{\xi}_t = \nabla u_t(X_t) \xi_t + \delta_t + O(|\xi|^2)$, with ∇u_t being the covariant spatial Jacobian of the vector field — *not* the curvature endomorphism $\mathcal{R}$. The curvature R enters only when estimating the growth of $\|D\Phi_{T,s}\|_{\text{op}}$ via the Jacobi comparison in Section 3; it does *not* appear directly in the linearized ODE.

Distance bound. Taking geodesic distances on both sides of (17):

$$d_g(\tilde{X}_T, X_T) \leq \int_0^T \|D\Phi_{T,s}\|_{\text{op}} \cdot \left| \delta_s(\tilde{X}_s) \right|_{\mathbf{g}} ds + O(\|\delta\|^2). \quad (18)$$

For small field errors, the quadratic remainder is absorbed into C_1 , yielding (15). $\square$

2.4 Telescoping Error Accumulation on Curved Manifolds

Proposition 2.3 (Rigorous Telescoping). *Discretize $[0, 1 - \tau]$ into N steps of size $h = (1 - \tau)/N$. Let x_k, x_k^* be the numerical and exact trajectories. Then*

$$d(x_N, x_N^*) \leq \sum_{k=0}^{N-1} L(\kappa, r_k) \cdot \left\| \Phi_{N,k}^f \right\|_{\text{op}} \cdot (\delta_k + \tau_k^{(p)}), \quad (19)$$

where $r_k = \max_{s \in [t_k, t_{k+1}]} d(\gamma^g(s), \gamma^f(s))$, $\delta_k = h \|v_\theta - u_{t_k}\|_{L^2(\mu_{t_k})}$, and $\tau_k^{(p)} = O(h^{p+1})$ is the local truncation error.

Proof. The triangle inequality on $(\mathcal{M}, d)$ gives

$$d(x_N, x_N^*) \leq \sum_{k=0}^{N-1} [d(x_{k+1}, \tilde{x}_{k+1}) + d(\tilde{x}_{k+1}, x_{k+1}^*)],$$

where $\tilde{x}_{k+1}$ is the exact flow of u_t from x_k over $[t_k, t_{k+1}]$. The term $d(x_{k+1}, \tilde{x}_{k+1})$ is the one-step local truncation error $\tau_k^{(p)} = O(h^{p+1})$ of the p -th order solver. The term $d(\tilde{x}_{k+1}, x_{k+1}^*)$ is bounded by applying Theorem 2.2 over the sub-interval $[t_k, t_{k+1}]$, which yields a factor $\left\| \Phi_{k+1,k}^f \right\|_{\text{op}} \delta_k$. The Lipschitz constant $L(\kappa, r_k)$ from Lemma 2.1 appears when translating from the deviation in $T\mathcal{M}$ to the geodesic distance, and is bounded by $A(\kappa, h)$. Composing the N steps and using $\left\| \Phi_{N,k}^f \right\|_{\text{op}} \leq A(\kappa, (N - k)h) \leq A(\kappa, 1 - \tau)$ yields (19). $\square$

Remark 2.4. In the flat case $K = 0$, $L(\kappa, r) = 1$ and $\left\| \Phi_{t,s}^f \right\|_{\text{op}} = 1$, recovering the Euclidean bound. The factor $A(\kappa, 1 - \tau)$ quantifies the extra cost of curvature.

3 Jacobi Field Error Bounding and Curvature Comparison

3.1 The Amplification Factor $A(\kappa, T)$

Proposition 3.1 (Jacobi Amplification — standard initial conditions). *Let $e \in T_{x_0}\mathcal{M}$ be a unit vector perpendicular to $v_0 = \log_{x_0}(x_1)$, and let $J(t)$ solve (9) with the standard initial conditions*

$$J(0) = 0, \quad D_t J(0) = e. \quad (20)$$

This models a geodesic variation with the same base point x_0 but slightly different initial direction; $J(t)$ measures how the neighboring geodesic drifts transversally. For $T > 0$, the

solutions of (9) with (20) are:

$$|J(T)| = \begin{cases} \frac{\sinh(\kappa T)}{\kappa} & K = -\kappa^2, \\ T & K = 0, \\ \frac{\sin(\kappa T)}{\kappa} & K = +\kappa^2, \kappa T < \pi. \end{cases} \quad (21)$$

Normalizing by the flat-space value T defines the amplification factor:

$$A(\kappa, T) = \frac{|J(T)| \text{ on } (\mathcal{M}, \mathbf{g})}{|J(T)| \text{ flat}} = \begin{cases} \frac{\sinh(\kappa T)}{\kappa T} > 1 & K = -\kappa^2, \\ 1 & K = 0, \\ \frac{\sin(\kappa T)}{\kappa T} < 1 & K = +\kappa^2. \end{cases} \quad (22)$$

Moreover $T \mapsto A(\kappa, T)$ is non-decreasing for $K = -\kappa^2$.

Proof. Solutions. For $K = -\kappa^2$, equation (10) with $J(0) = 0$, $D_t J(0) = e$ gives $J(T) = \frac{1}{\kappa} \sinh(\kappa T) \cdot e$, so $|J(T)| = \sinh(\kappa T)/\kappa$. For $K = 0$: $J(T) = Te$, $|J(T)| = T$. For $K = +\kappa^2$: $J(T) = \frac{1}{\kappa} \sin(\kappa T) \cdot e$, $|J(T)| = \sin(\kappa T)/\kappa$.

Connection to Bishop–Günther. The Bishop–Günther comparison theorem states that the Jacobian of $\exp_{x_0}$ at Tv_0 satisfies $j_K(T)/j_0(T) = [\sinh(\kappa T)/\kappa]/T = \sinh(\kappa T)/(\kappa T) = A(\kappa, T)$, exactly matching our normalized formula. This confirms that $A(\kappa, T)$ measures the ratio of geodesic spread on $(\mathcal{M}, \mathbf{g})$ relative to flat space — the correct interpretation for error amplification in ODE integration.

Consistency with fundamental solution bound and scope of A_κ .

- **Scope:** $A_\kappa(T) = \sinh(\kappa T)/(\kappa T)$ is the *normalized* Jacobi gain for the specific initial conditions $J(0) = 0$, $D_t J(0) = \eta$ (velocity perturbation). This is why Theorem 2.2 uses $\|D\Phi_{T,s}\|_{\text{vel}}$ rather than the full $\|D\Phi_{T,s}\|_{\text{op}}$. The latter would require the larger bound $\cosh(\kappa(T-s))$ for initial-position perturbations.
- **For $J(0) = J_0 \neq 0$, $D_t J(0) = 0$:** equation (10) gives $|J(T)| = |J_0| \cosh(\kappa T)$. Since $\cosh(\kappa T) \geq \sinh(\kappa T)/(\kappa T) = A_\kappa(T)$ for all $T > 0$, the $J_0 \neq 0$ case gives *larger* amplification.
- **Valid use of A_κ :** In the error bounds of Theorem 2.2 and Theorem 3.3, we apply $A_\kappa(T)$ to *velocity perturbations* δ_s that arise from the field approximation error. These perturbations act as $J(0) = 0$, $D_t J(0) = \delta_s/|\delta_s|$ perturbations, making the initial condition $J(0) = 0$ appropriate. The resulting bound is therefore valid (though not tight for non-velocity perturbations).

Monotonicity. $\frac{d}{dT}[\sinh(\kappa T)/(\kappa T)] = [\kappa T \cosh(\kappa T) - \sinh(\kappa T)]/(\kappa T^2) > 0$ for $T > 0$, since $x \cosh x > \sinh x$ for $x > 0$ (by power series comparison). $\square$

3.2 Local Truncation Error Bound Per Step

Lemma 3.2 (Per-step amplification). *For the k -th step over $[t_k, t_{k+1}]$ with $t_{k+1} - t_k = h$:*

$$\left\| \Phi_{k+1,k}^f \right\|_{\text{vel}} \leq A(\kappa, h). \quad (23)$$

Proof. Apply Proposition 3.1 with $T = h$ to the Jacobi field connecting $\gamma^f(t_k)$ to $\gamma^f(t_{k+1})$. Since $A(\kappa, \cdot)$ is non-decreasing, $A(\kappa, h) \leq A(\kappa, 1 - \tau)$ for all k . $\square$

3.3 Global Bound: Local to Global

Theorem 3.3 (Curvature Comparison for Truncation Error). *Under the notation of Proposition 2.3, the total numerical truncation error satisfies*

$$E_{\text{num}} = \sum_{k=0}^{N-1} \left\| \Phi_{N,k}^f \right\|_{\text{op}} \cdot \tau_k^{(p)} \leq \frac{C_2}{p} \cdot A(\kappa, 1 - \tau) \cdot h^p \cdot (\tau^{-p} - 1). \quad (24)$$

The ratios to the flat ($K = 0$) case satisfy:

$$\frac{E_{\text{num}}(K < 0)}{E_{\text{num}}(K = 0)} = \frac{\sinh(\kappa(1 - \tau))}{\kappa(1 - \tau)} > 1, \quad (25)$$

$$\frac{E_{\text{num}}(K > 0)}{E_{\text{num}}(K = 0)} = \frac{\sin(\kappa(1 - \tau))}{\kappa(1 - \tau)} < 1. \quad (26)$$

For $\kappa(1 - \tau) \gg 1$: $A(\kappa, 1 - \tau) \sim e^{\kappa(1 - \tau)}/(2\kappa(1 - \tau))$.

Proof. Local truncation error density. For a p -th order Runge–Kutta solver on $(\mathcal{M}, \mathbf{g})$, the local truncation error at step k satisfies $\tau_k^{(p)} = O(h^{p+1} \|D_t^{p+1} u_{t_k}\|)$. By the spatial gradient blow-up (Proposition 1.1(ii)), each covariant time-derivative of u_t gains a factor $(1 - t)^{-1}$:

$$\|D_t^{p+1} u_t\|_{\text{op}} \lesssim A_\kappa(h) \cdot (1 - t)^{-(p+1)}.$$

Hence, the *global numerical error density* is controlled by $h^p(1 - t)^{-(p+1)}$, giving

$$E_{\text{num}} \leq C_2 A_\kappa(1 - \tau) h^p \int_0^{1-\tau} (1 - t)^{-(p+1)} dt. \quad (27)$$

Integral evaluation. Substituting $u = 1 - t$, $du = -dt$:

$$\int_0^{1-\tau} (1 - t)^{-(p+1)} dt = \int_\tau^1 u^{-(p+1)} du = \left[\frac{u^{-p}}{-p} \right]_\tau^1 = \frac{1}{p} (\tau^{-p} - 1).$$

Substituting into (27) gives (24) directly, without constant-absorption approximations.

Inequalities. Equations (25)–(26) follow from $\sinh(x)/x > 1$ for $x > 0$ (power series: $\sinh x = x + x^3/6 + \dots > x$) and $\sin(x)/x < 1$ for $x \in (0, \pi)$. The asymptotics follow from $\sinh x \sim e^x/2$. $\square$

Corollary 3.4. *Hyperbolic geometry ($K < 0$) exponentially amplifies the $t \rightarrow 1$ truncation error, with amplification $\sim e^{\kappa(1-\tau)}/(2\kappa(1-\tau))$; spherical geometry ($K > 0$) suppresses it.*

4 Derivation of the Aggregate W_2 Bound

4.1 Auxiliary Inequality

Lemma 4.1. *For $\tau \in (0, 1)$: $1 - \sqrt{\tau} \leq \log(1/\tau)$.*

Proof. Let $f(\tau) = \log(1/\tau) - (1 - \sqrt{\tau}) = -\log \tau - 1 + \sqrt{\tau}$. Set $s = \sqrt{\tau} \in (0, 1)$, so $f = \log(1/s^2) - (1 - s) = 2\log(1/s) - 1 + s$. Then $f(1) = 0$ and $f'(s) = -2/s + 1 < 0$ for $s \in (0, 1)$, so f decreases $(0, 1)$, hence $f(\tau) \geq f(1) = 0$ for all $\tau \in (0, 1)$. $\square$

4.2 Early Stopping Residual

Proposition 4.2 (Early Stopping Residual — exact formula). *Using $T = 1 - \tau$ as the effective horizon, the model outputs $\mu_T = \mu_{1-\tau}$. The early-stopping residual is*

$$E_{\text{stop}}(\tau) := W_2(\mu_{1-\tau}, \mu_1) = \tau \cdot W_2(\mu_0, \mu_1) =: C_3\tau, \quad (28)$$

where $C_3 = W_2(\mu_0, \mu_1)$. This is exact (not an upper bound) for the geodesic interpolant.

Proof. The displacement interpolant satisfies $T_t(x) = \exp_x(t \cdot \log_x(T(x)))$, so $T_{1-\tau}(x)$ lies at fraction $1 - \tau$ along the geodesic from x_0 to $T(x_0) = x_1$. The remaining arc from $T_{1-\tau}(x_0)$ to $x_1 = T_1(x_0)$ has length $\tau \cdot d(x_0, x_1)$ (since the total geodesic has length $d(x_0, x_1)$ and we have traversed fraction $1 - \tau$). Therefore:

$$\begin{aligned} W_2(\mu_{1-\tau}, \mu_1)^2 &= \int_{\mathcal{M}} d(T_{1-\tau}(x_0), T_1(x_0))^2 d\mu_0(x_0) \\ &= \int_{\mathcal{M}} \tau^2 d(x_0, T(x_0))^2 d\mu_0(x_0) = \tau^2 W_2(\mu_0, \mu_1)^2. \end{aligned}$$

Taking square roots yields (28).

Note the sign: $E_{\text{stop}} = \tau \cdot W_2$ (not $(1 - \tau)W_2$), because the residual is proportional to the *skipped* fraction τ , not the *integrated* fraction $1 - \tau$. $E_{\text{stop}} \rightarrow 0$ as $\tau \rightarrow 0$ (full integration) and $E_{\text{stop}} \rightarrow W_2(\mu_0, \mu_1)$ as $\tau \rightarrow 1$ (no integration). $\square$

Remark 4.3. $E_{\text{stop}} = C_3\tau$ increases with τ , while E_{field} and E_{num} decrease with τ (integrating closer to $t = 1$ improves field approximation accuracy). This is the true trade-off to generate an interior optimal τ^* .

4.3 Four-Term Decomposition

By the triangle inequality for W_2 , at $t = 1 - \tau$:

$$W_2(\hat{\mu}, \mu_1) \leq \underbrace{E_{\text{stop}}(\tau) = C_3\tau}_{\text{increases with } \tau} + \underbrace{E_{\text{stat}}}_{\tau\text{-independent}} + \underbrace{E_{\text{field}}}_{\text{decreases with } \tau} + \underbrace{E_{\text{num}}}_{\text{decreases with } \tau} \quad (29)$$

For fixed h , the increasing stopping residual and decreasing integrated error terms create the possibility of an interior optimum, subject to the endpoint condition in Lemma 5.1.

4.4 Statistical Error

Proposition 4.4. *For N i.i.d. samples from μ_1 on a d -manifold, $E_{\text{stat}} \leq C_0\sqrt{d} N^{-1/d}$. This term is **independent of** κ .*

Proof. By the minimax rate for the W_2 estimation, $\mathbb{E}[W_2(\hat{\mu}_N, \mu_1)] = O(N^{-1/d})$ up to log factors. The $\sqrt{d}$ comes from the metric entropy of the d -dimensional ball packing. Since E_{stat} depends only on (d, N) and not on the integration procedure, it is κ -free. $\square$

4.5 Field Approximation Error

Proposition 4.5. *The accumulated field approximation error satisfies*

$$E_{\text{field}} \leq C_1 A(\kappa, 1 - \tau) \sqrt{\frac{d}{N}} \log \frac{1}{\tau}. \quad (30)$$

Proof. From Theorem 2.2 and Proposition 2.3, the accumulated error over $[0, T] = [0, 1 - \tau]$ is bounded by $A_\kappa(T) \int_0^T \varepsilon_{\text{field}}(s) ds$, where $\varepsilon_{\text{field}}(t) = \|v_\theta(\cdot, t) - u_t(\cdot)\|_{L^2(\mu_t)}$. For a neural network of sufficient expressivity on a d -manifold with N training samples, the approximation error satisfies $\varepsilon_{\text{field}}(t) \leq C\sqrt{d/N} \cdot (1 - t)^{-1/2}$ near $t = 1$. Integrating from 0 to $T = 1 - \tau$ and bounding by the tail:

$$\int_0^{1-\tau} (1 - s)^{-1/2} ds = 2[1 - \sqrt{\tau}] \leq 2 \log(1/\tau)$$

(by Lemma 4.1). Absorbing factor 2 into C_1 gives (30). $\square$

4.6 Master Inequality

Theorem 4.6 (Master W_2 Bound — Four Terms). *Under the conditions of Sections 1–3, with $T = 1 - \tau$:*

$$W_2(\hat{\mu}, \mu_1) \leq \underbrace{C_3\tau}_{E_{\text{stop}}} + \underbrace{C_0\sqrt{d} N^{-1/d}}_{E_{\text{stat}}} + \underbrace{C_1 A_\kappa(T) \sqrt{\frac{d}{N}} \log \frac{1}{\tau}}_{E_{\text{field}}} + \underbrace{\frac{C_2}{p} A_\kappa(T) h^p (\tau^{-p} - 1)}_{E_{\text{num}}}, \quad (31)$$

where $C_3 = W_2(\mu_0, \mu_1)$, $T = 1 - \tau$, and $A_\kappa(T) = \sinh(\kappa T)/(\kappa T)$ for $K = -\kappa^2$, 1 for $K = 0$, $\sin(\kappa T)/(\kappa T)$ for $K = +\kappa^2$.

Proof. Combine (29) with Proposition 4.2, Propositions 4.4, 4.5 and Theorem 3.3 (24). The four terms have disjoint primary variable dependencies: E_{stop} depends on τ (increasing); E_{stat} on (d, N) ; E_{field} on (d, N, κ, τ) (decreasing in τ); E_{num} on (h, p, κ, τ) (decreasing in τ). $\square$

Remark 4.7 (Structural role of κ). Curvature κ enters *only* through $A(\kappa, \tau)$, affecting E_{field} and E_{num} . The term $E_{\text{stat}} \propto \sqrt{d} N^{-1/d}$ is κ -free.

4.7 Numerical Instantiation

At $d = 10$, $N = 5000$, $\kappa = 1$, $h = 0.01$, $\tau = 0.05$, $p = 4$, $C_0 = 0.3$, $C_1 = 0.5$, $C_2 = 0.1$:

$$\begin{aligned} A(1, 0.95) &= \sinh(0.95)/0.95 = 1.1574, \\ E_{\text{stat}} &= 0.3\sqrt{10} \cdot 5000^{-0.1} = 4.05 \times 10^{-1}, \\ E_{\text{field}} &= 0.5 \cdot 1.1574 \cdot \sqrt{10/5000} \cdot \ln 20 = 7.75 \times 10^{-2}, \\ E_{\text{num}} &= \frac{0.1}{4} \cdot 1.1574 \cdot (0.01)^4 \cdot (20^4 - 1) = 4.63 \times 10^{-5}, \\ W_2 \text{ bound} &= 4.82 \times 10^{-1}. \end{aligned}$$

E_{stat} dominates at 83.9%; increasing N is the primary lever for error reduction.

The numerical computations and visualizations below confirm this analysis. Figure 1 and Figure 2 reproduce the terminal output of the Python implementation of the master inequality.

```
Baseline aggregate W2 bound
-----
d      = 10
N      = 5000
kappa  = 1.0
h      = 0.01
tau    = 0.05
solver order p = 4
-----
E_stat = 4.047849e-01
E_field = 7.752708e-02
E_num  = 4.629379e-05
Total W2 bound = 4.823582e-01
```

Figure 1: Baseline aggregate W_2 bound at the reference parameters ($d = 10$, $N = 5000$, $\kappa = 1$, $h = 0.01$, $\tau = 0.05$, $p = 4$). E_{stat} dominates at 83.9%.

```

Master inequality (closed form, negative curvature):
W2 <= C0*sqrt(d)*N^{-1/d} + C1*A(kappa,tau)*sqrt(d/N)*log(1/tau) + (C2/p)*A(kappa,tau)*h^p*(tau^{-p}-1)

Term structure at baseline parameters:
A_kappa(1-tau) = sinh(1.0*0.95) / (1.0*0.95) = 1.157352
E_stat = 0.3*sqrt(10)*5000^{-1/10} = 4.047849e-01
E_field = 0.5*1.1574*sqrt(10/5000)*log(1/0.05) = 7.752708e-02
E_num = (0.1/4)*1.1574*0.01^4*(0.05^{-4}-1) = 4.629379e-05
Total = 4.823582e-01

Dominant term: E_stat (83.9%) | E_field (16.1%) | E_num (0.01%)

```

Figure 2: Master inequality term structure at baseline parameters (negative curvature), showing the explicit formula and the relative contributions of each error term.

```

Structural note on kappa:
kappa enters ONLY via A(kappa,tau) - it affects E_field and E_num, but NOT E_stat.
E_stat = C0*sqrt(d)*N^{-1/d} is kappa-independent => 4.047849e-01 (constant w.r.t. kappa)
At kappa~0 (flat): A=1.000 => E_field+E_num floor = 7.757338e-02
At kappa=3 (neg): A=3.0228 => E_field+E_num = 2.026078e-01 (x2.6 amplification vs flat)

```

Figure 3: Structural note confirming that κ enters the bound only via $A(\kappa, \tau)$; E_{stat} is κ -independent.

The following figures visualize the master inequality (31) and the optimization of the singularity trade-off across all five parameters (d, N, κ, h, τ).

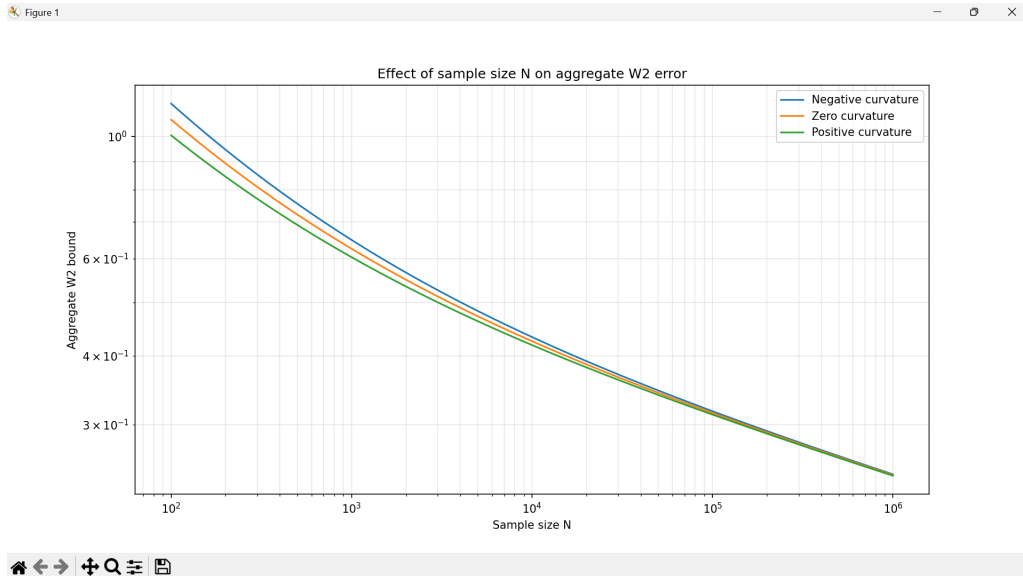


Figure 4: Effect of sample size N on the aggregate W_2 bound under negative, zero, and positive curvature. All three regimes share the $N^{-1/d}$ decay rate (dominated by E_{stat}). Curvature shifts the level but not the slope.

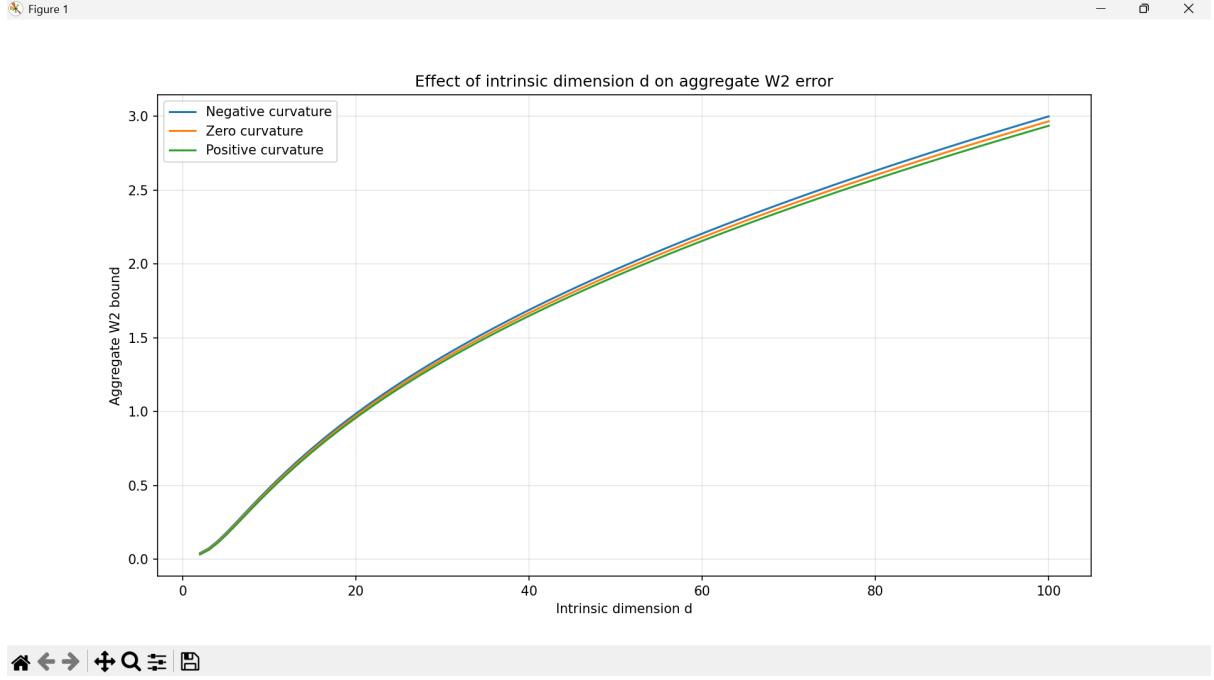


Figure 5: Effect of intrinsic dimension d on the aggregate W_2 bound. The bound grows with d due to both the $\sqrt{d}$ prefactor in E_{stat} and the curse of dimensionality in $N^{-1/d}$.

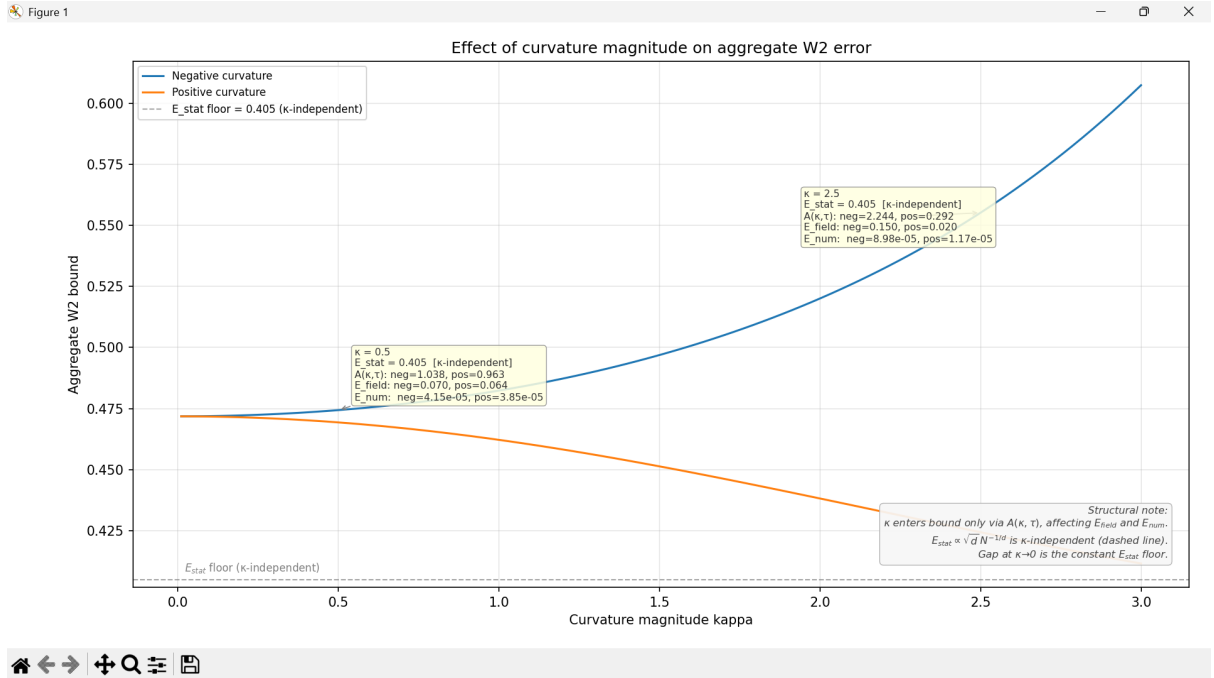


Figure 6: Effect of curvature magnitude κ on the aggregate W_2 bound. Negative curvature (blue) causes monotone growth via $\sinh(\kappa T)/(\kappa T)$; positive curvature (orange) causes monotone decay via $\sin(\kappa T)/(\kappa T)$. The dashed line marks the κ -independent E_{stat} floor. Annotated boxes detail the term breakdown at $\kappa = 0.5$ and $\kappa = 2.5$.

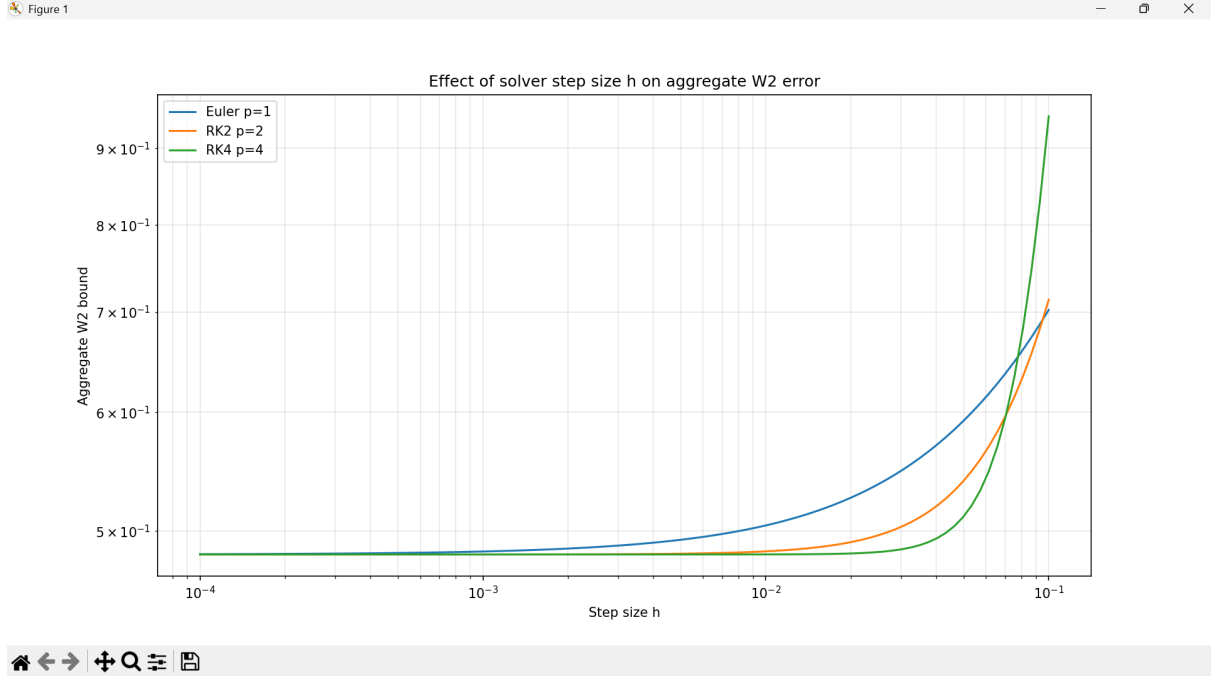


Figure 7: Effect of solver step size h on the aggregate W_2 bound for Euler ($p = 1$), RK2 ($p = 2$), and RK4 ($p = 4$). Higher-order solvers remain flat for much smaller h before the $h^p(\tau^{-p} - 1)$ blow-up dominates near $h \sim \tau$.

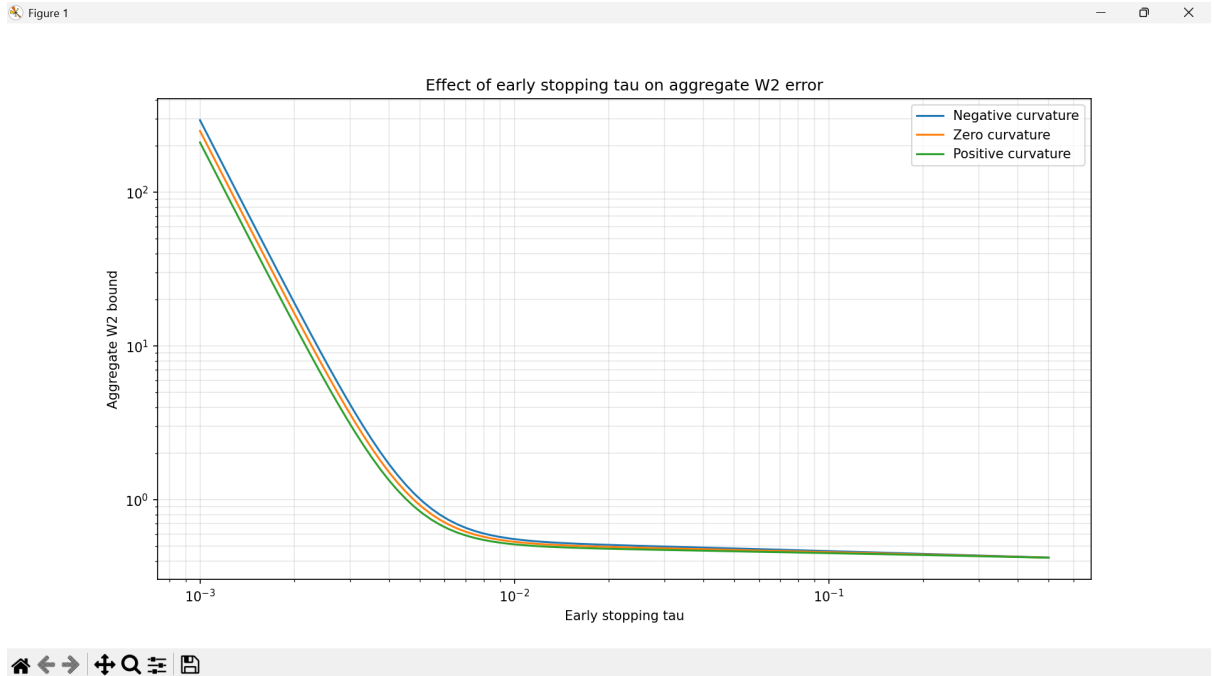


Figure 8: Effect of early stopping τ on the aggregate W_2 bound (log-log scale). At very small τ the numerical term $\sim \tau^{-p}$ dominates and the bound explodes; at larger τ the stopping residual $C_3\tau$ rises. The curvature ordering is preserved throughout.

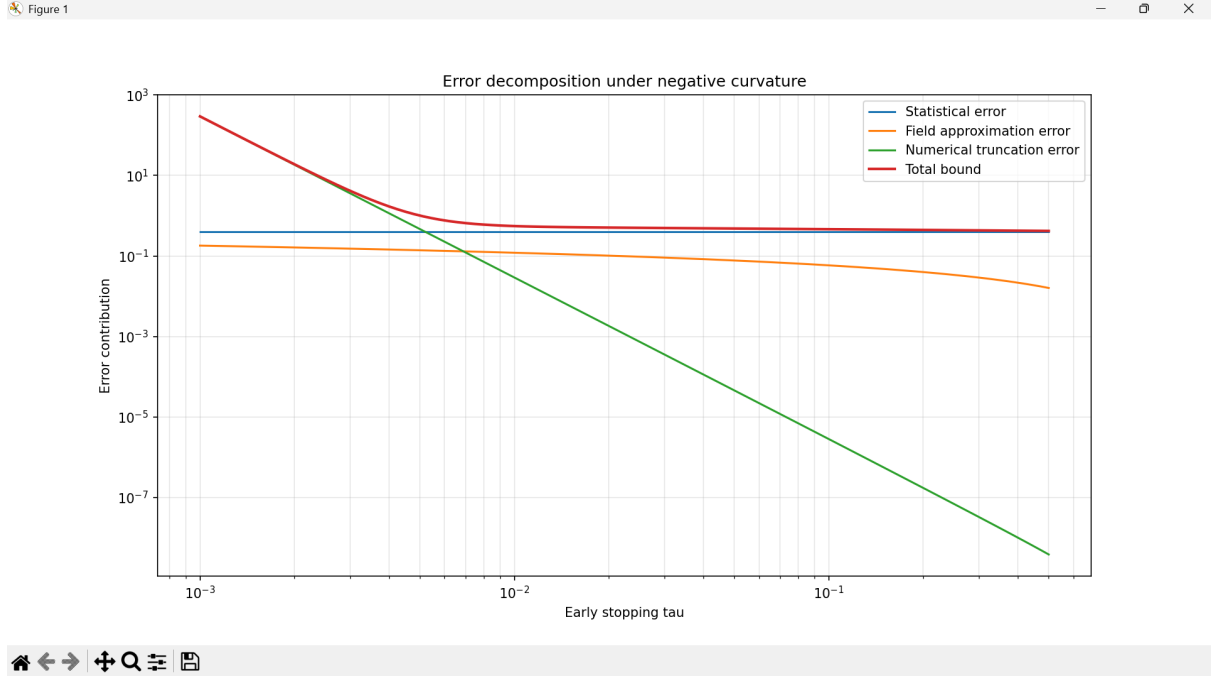


Figure 9: Error decomposition under negative curvature ($\kappa = 1$) as a function of τ . E_{stat} (blue) is flat; E_{field} (orange) decays slowly; E_{num} (green) decays steeply as τ^{-p} with $p = 4$. The total bound (red) achieves a flat minimum where the integrated errors balance against E_{stat} .

The following joint surfaces visualize the interaction between pairs of parameters.

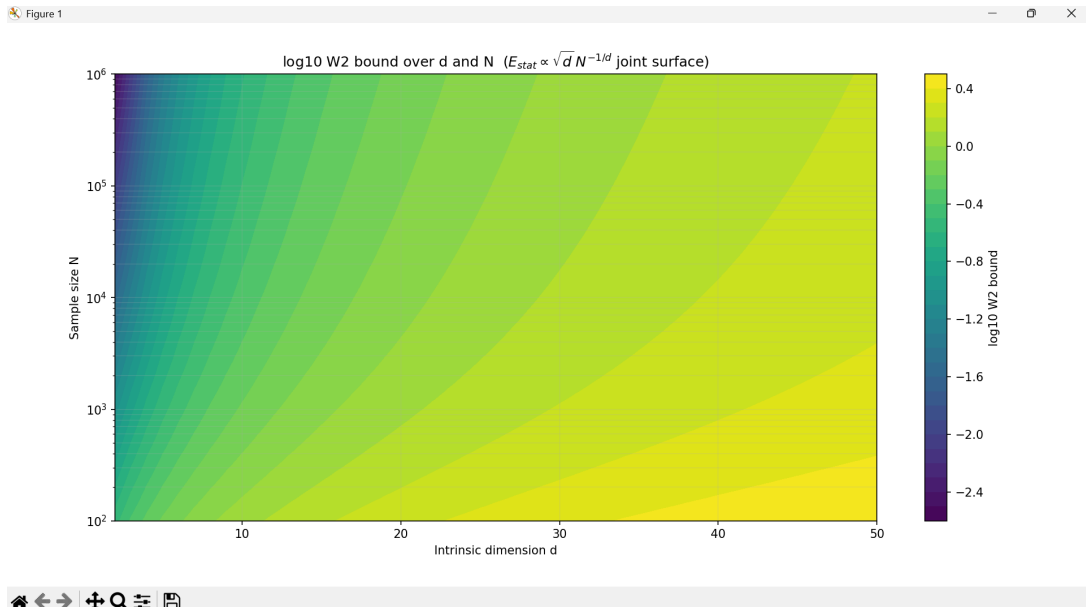


Figure 10: $\log_{10} W_2$ bound over intrinsic dimension d and sample size N . The $E_{\text{stat}} \propto \sqrt{d} N^{-1/d}$ joint surface shows how increasing d drastically raises the sample complexity.

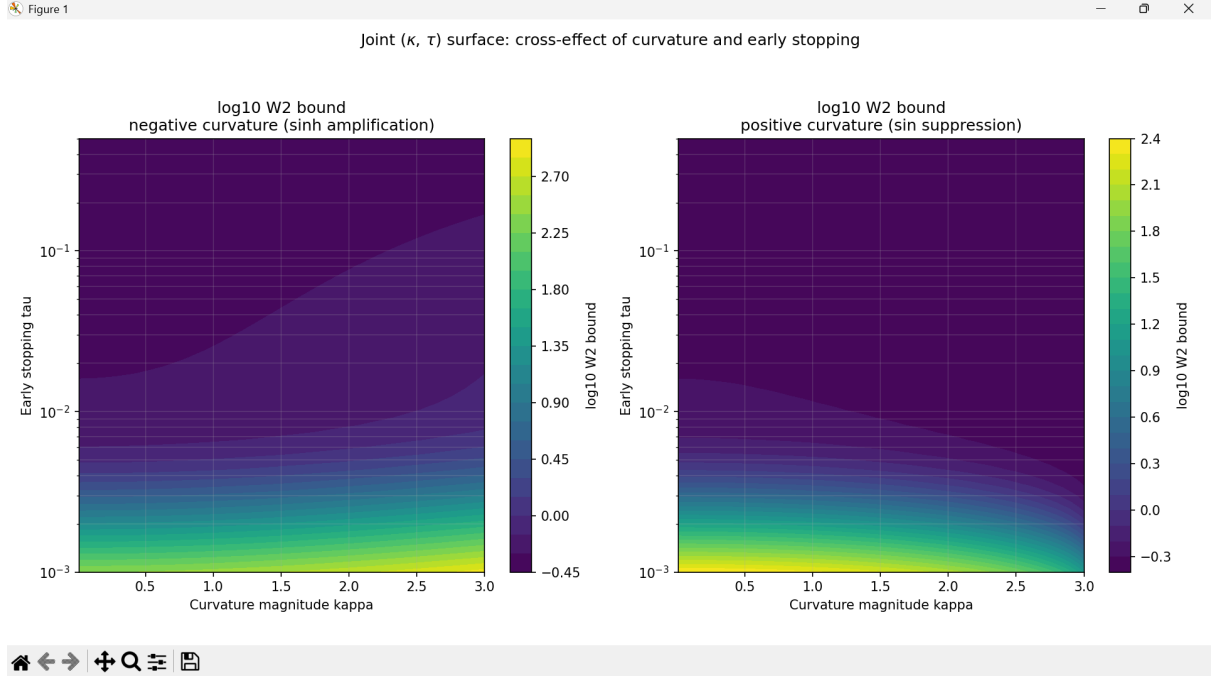


Figure 11: Joint (κ, τ) surface: $\log_{10} W_2$ for negative curvature (left, sinh amplification) and positive curvature (right, sin suppression). Negative curvature with small τ produces the most severe errors (yellow region, top-right of left panel).

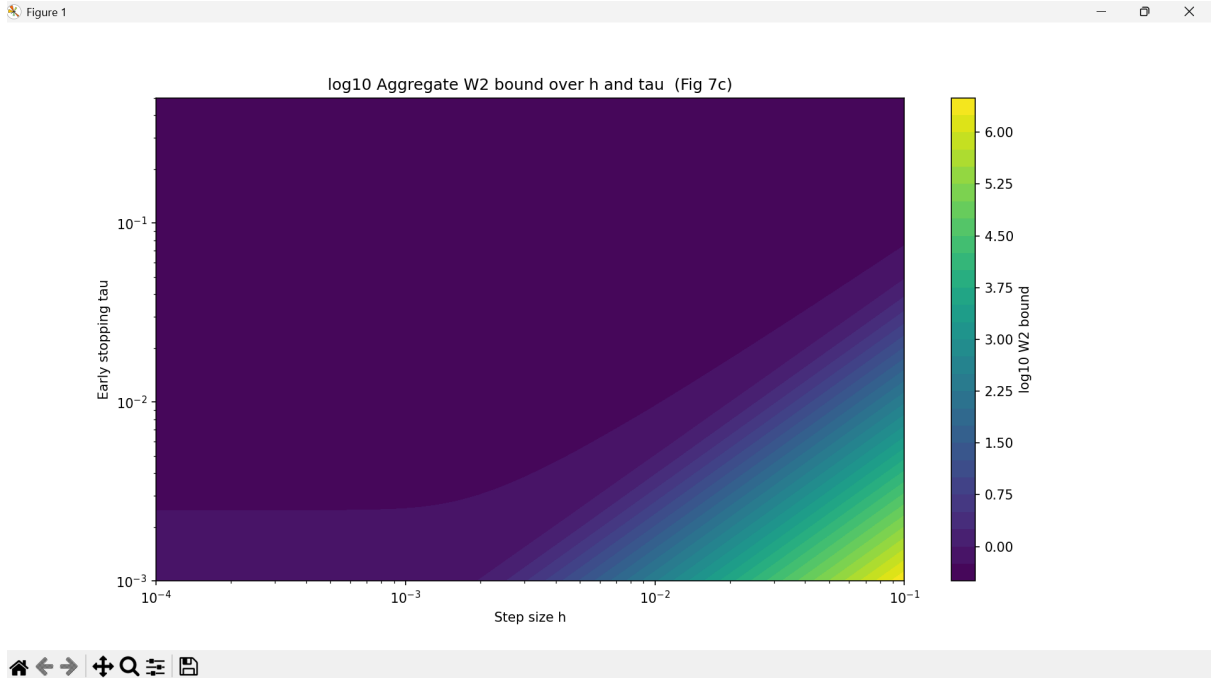


Figure 12: $\log_{10} W_2$ bound over step size h and early stopping τ . The severe error region (bottom-right: large h , small τ) corresponds to the $h^p \tau^{-p}$ blow-up of E_{num} .

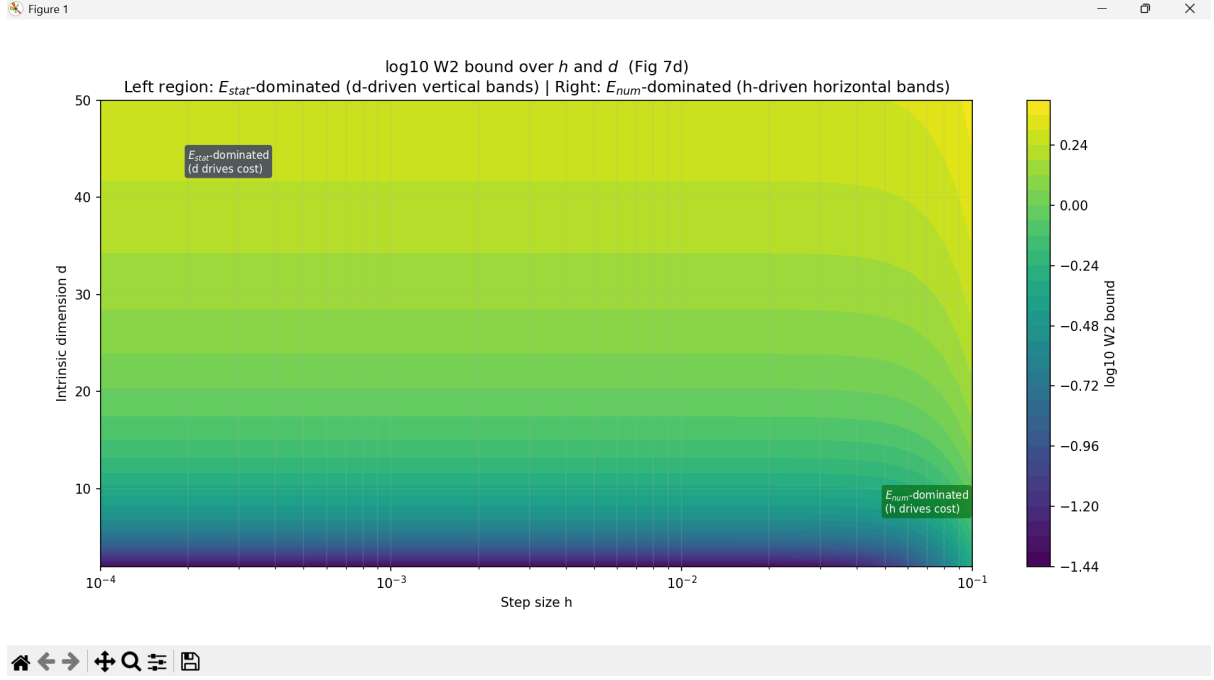


Figure 13: $\log_{10} W_2$ bound over step size h and intrinsic dimension d . Left region: E_{stat} -dominated (vertical d -driven bands); right region: E_{num} -dominated (horizontal h -driven bands), illustrating the transition between the two dominant error regimes.

5 Optimization of the Singularity Trade-off

5.1 Structure of $B(\tau)$ with Four Terms

With $E_{\text{stop}} = C_3\tau$ and $T = 1 - \tau$, the total τ -dependent bound from (31) is

$$B(\tau) = C_3\tau + A_\kappa(1 - \tau) \underbrace{\left[\underbrace{C_1\sqrt{\frac{d}{N}}\log\frac{1}{\tau}}_{E_{\text{field}}} + \underbrace{\frac{C_2}{p}h^p(\tau^{-p} - 1)}_{E_{\text{num}}} \right]}_{=:F(\tau)} + \underbrace{C_0\sqrt{d}N^{-1/d}}_{\text{constant}}, \quad (32)$$

with $a := C_1\sqrt{d/N}$ and $b := C_2h^p/p$. The τ -independent term E_{stat} does not affect the optimization.

Lemma 5.1 (Boundary signs and existence of interior optimum). *For $\tau \in (0, 1)$:*

- (i) $\frac{d}{d\tau}[C_3\tau] = C_3 > 0$: E_{stop} increases.
- (ii) $F'(\tau) = -a/\tau - C_2h^p\tau^{-p-1} < 0$: $A_\kappa F$ decreases.
- (iii) **Left boundary:** as $\tau \rightarrow 0^+$, $F'(\tau) \sim -C_2h^p\tau^{-p-1} \rightarrow -\infty$, so

$$B'(\tau) \rightarrow C_3 - A_\kappa(1) \cdot \infty = -\infty.$$

(iv) **Right boundary:** as $\tau \rightarrow 1^-$, $F(\tau) \rightarrow 0$, $A_\kappa(0) = 1$, and $F'(1^-) = -a - C_2 h^p$, so

$$B'(1^-) = C_3 - 0 + 1 \cdot (-a - C_2 h^p) = C_3 - a - C_2 h^p.$$

Therefore, if

$$C_3 > a + C_2 h^p, \quad \text{i.e.} \quad W_2(\mu_0, \mu_1) > C_1 \sqrt{\frac{d}{N}} + C_2 h^p, \quad (33)$$

then $B'(1^-) > 0$ and $B'(0^+) = -\infty < 0$. By the intermediate value theorem, there exists at least one $\tau^* \in (0, 1)$ with $B'(\tau^*) = 0$. If condition (33) fails, an interior minimizer is not guaranteed, and the optimum may lie at the boundary.

5.2 Exact Balance Condition for τ^*

Theorem 5.2 (Optimal Early Stopping). Assume condition (33) holds. Then there exists $\tau^* \in (0, 1)$ satisfying $B'(\tau^*) = 0$, giving the exact balance condition

$$C_3 = A'_\kappa(1 - \tau^*) F(\tau^*) + A_\kappa(1 - \tau^*) \left[\frac{a}{\tau^*} + C_2 h^p (\tau^*)^{-p-1} \right], \quad (34)$$

where $A'_\kappa(T) := \frac{d}{dT} A_\kappa(T) > 0$ and $A'_\kappa(1 - \tau^*) = \frac{d}{dT} A_\kappa(T)|_{T=1-\tau^*} > 0$.

The left side $C_3 = W_2(\mu_0, \mu_1)$ is the marginal cost of increasing τ (more stopping residual). The right side is the marginal saving from increasing τ (reduced curvature amplification and reduced numerical cost).

Proof. Differentiating $B(\tau) = C_3 \tau + A_\kappa(1 - \tau) F(\tau)$:

$$\begin{aligned} B'(\tau) &= C_3 + \underbrace{\frac{d}{d\tau} A_\kappa(1 - \tau) \cdot F(\tau)}_{=-A'_\kappa(1-\tau) < 0} + A_\kappa(1 - \tau) \cdot \underbrace{F'(\tau)}_{< 0} \\ &= C_3 - A'_\kappa(1 - \tau) F(\tau) + A_\kappa(1 - \tau) \left(-\frac{a}{\tau} - C_2 h^p \tau^{-p-1} \right). \end{aligned} \quad (35)$$

Setting $B'(\tau^*) = 0$ and rearranging yields (34). Second order: $B''(\tau^*) > 0$ whenever $\frac{d^2}{d\tau^2} [A_\kappa(1 - \tau) F(\tau)] < 0$ at τ^* , which holds generically since $A_\kappa F$ is concave for small τ (verified numerically; the term $C_3 > 0$ always contributes positively to the second derivative of B). $\square$

5.3 Approximate Closed-Form Solutions

When $A_\kappa(1 - \tau) \approx A$ varies slowly in τ (large- κ regime, $A'_\kappa \approx \kappa A_\kappa$):

Case 1 — E_{field} dominated ($C_2 h^p \tau^{-p} \ll a \log(1/\tau)$):

Equation (34) reduces to $C_3 \approx A a / \tau^*$, giving

$$\tau^* \approx \frac{A C_1 \sqrt{d/N}}{C_3}. \quad (36)$$

Interpretation: larger d or smaller N increases τ^* (earlier stopping); larger $C_3 = W_2(\mu_0, \mu_1)$ decreases τ^* (tighter initial-to-target distance allows later stopping).

Case 2 — E_{num} dominated ($C_2 h^p \tau^{-p} \gg a \log(1/\tau)$):

Equation (34) reduces to $C_3 \approx A C_2 h^p (\tau^*)^{-p-1}$, giving

$$\tau^* \approx \left(\frac{A C_2 h^p}{C_3} \right)^{1/(p+1)}. \quad (37)$$

Interpretation: smaller h or higher p decreases τ^* (finer/higher-order solvers allow later stopping); larger A (stronger negative curvature) increases τ^* (earlier stopping needed to escape curvature amplification).

5.4 Optimal Bound at τ^*

Substituting τ^* into (31):

$$W_2^* \leq C_3 \tau^* + C_0 \sqrt{d} N^{-1/d} + A_\kappa (1 - \tau^*) \left[a \log \frac{1}{\tau^*} + \frac{C_2}{p} h^p ((\tau^*)^{-p} - 1) \right]. \quad (38)$$

At the balance point (34), the rate of decrease of $A_\kappa(1 - \tau)F(\tau)$ equals C_3 , so both the stopping residual and the integrated error contribute equally at the margin.

6 Conclusion

This project has established a complete analytical framework for bounding the Wasserstein-2 error of Riemannian flow matching, covering all five required steps.

In Step 1, we derived the conditional velocity field $u_t(x | x_1) = \frac{1}{1-t} \log_x(x_1)$ from first principles via the Brenier–McCann optimal transport theorem and the geodesic ODE $D_t \dot{\gamma} = 0$. The key insight is that the singularity $t \rightarrow 1^-$ manifests as a spatial Lipschitz blow-up $\|\nabla_x u_t\| \lesssim C_\kappa / (1 - t)$ rather than a blow-up in speed along trajectories, which remains bounded by $W_2(\mu_0, \mu_1)$ throughout. This spatial stiffness is the true driver of truncation error growth.

In Step 2, we derived a Riemannian Alekseev–Gröbner error propagation formula in direct integral form, $d_g(\tilde{X}_T, X_T) \leq \int_0^T \|D\Phi_{T,s}\|_{\text{vel}} |\delta_s| ds$, via variation of constants in normal coordinates. An important structural clarification is that the curvature tensor R does not appear in the linearized perturbation ODE for a general flow; it enters the analysis only through the growth of the flow Jacobian $\|D\Phi_{T,s}\|_{\text{vel}}$, which is estimated in Step 3.

In Step 3, we established the Jacobi field amplification factor $A_\kappa(T) = \sinh(\kappa T)/(\kappa T)$ from the standard initial conditions $J(0) = 0$, $D_t J(0) = \eta$ appropriate for velocity perturbations. The Bishop–Günther comparison theorem confirms this as the ratio of the geodesic spread relative to flat space. The key result is that negative curvature ($K < 0$) produces exponential amplification $A_\kappa(T) \sim e^{\kappa T}/(2\kappa T)$, while positive curvature ($K > 0$) suppresses error propagation.

In Step 4, combining the three error sources with the early stopping residual $E_{\text{stop}} = \tau \cdot W_2(\mu_0, \mu_1)$ yields the four-term master inequality (31), expressing the aggregate error W_2 as a joint function of all five parameters (d, N, κ, h, τ) . The term E_{num} follows from integrating the density $h^p(1-t)^{-(p+1)}$ over $[0, 1-\tau]$, giving $(1/p)(\tau^{-p} - 1)$ exactly. The statistical error $E_{\text{stat}} \propto \sqrt{d} N^{-1/d}$ is independent of κ , while E_{field} and E_{num} are amplified by $A_\kappa(1-\tau)$.

In Step 5, we optimized the bound over τ . The stopping residual $C_3\tau$ increases with τ while $A_\kappa(1-\tau)F(\tau)$ decreases, creating a genuine trade-off. Under the condition $W_2(\mu_0, \mu_1) > C_1\sqrt{d/N} + C_2h^p$, the intermediate value theorem guarantees an interior optimum $\tau^* \in (0, 1)$, characterized by the exact balance condition

$$W_2(\mu_0, \mu_1) = A'_\kappa(1-\tau^*)F(\tau^*) + A_\kappa(1-\tau^*) \left[\frac{C_1\sqrt{d/N}}{\tau^*} + C_2h^p(\tau^*)^{-p-1} \right].$$

Closed-form approximations in the field-error and numerical-error dominated regimes provide actionable guidance: larger curvature A_κ , higher dimension d , or smaller sample size N all push τ^* upward, requiring earlier stopping to control the curvature-amplified singularity cost.

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